



Camera Magic

A compilation showcasing special effects and film illusion techniques used in the 1940s film industry. <http://dgit.in/feb21-27>



Optical Printing and Blue Screens

A documentary highlighting ILM's special effects work on Star Wars Episode V: The Empire Strikes Back. <http://dgit.in/feb21-28>

makes a mark on them. This indication would then be used as the basis for cutting and cello taping shots together, using a small contraption known as a film splicer. For the next 50 to 60 years, not a great deal of innovation took place, besides the development of flatbed editors, which made things easier and faster by adding a giant display screen and a neat, robust workplace. But again, it followed the same base editing procedure. That is until digital film editing came into play.

VISUAL EFFECTS

Today, the term “Visual Effects” is impossible to think about, without directly associating it to computers and a whole bunch of software. But for the longest time, filmmakers used the most practical methods to create and integrate imageries that would add to the visual cinematic experience. After finalising the edits in the previous stage, it was now time to implement effects and bring the film to life. A commonly-used device at the time was an optical printer, which is a simple idea. The apparatus consists of a camera pointed at a film projector, lens to lens, and is designed to re-photograph film and add innumerable visual effects to a movie. Think of it as physically recording a running video off of a screen, which creates a near-perfect, fresh copy. A

filmmaker can then manipulate the image as it is being recorded, by slightly moving the camera around or by zooming their lenses in and out. Oftentimes, they also employed filters in between lenses to change colours or to introduce blurs. More advanced effects such as fade-ins and fade-outs were done by simply messing with the exposure, while dissolve transitions were achieved by timely recording over previously shot negatives.

Coming to the large-scale special-effects region, filmmakers would hire skilled artists to paint beautiful, realistic representations of artefacts, landscapes, sets or distant locations for their movies. This is done to create the illusion of an environment that is not present at the filming location. These matte paintings, at the time, were done on a sheet of glass and were mounted in front of the camera, before hitting record. This method was also used for covering up any unwanted elements within the framework.



Optical Printer



Compositing multiple layers of film over blue screen

Much later, blue screens made an appearance, letting the VFX team throw in any backdrop they desired. For this, the assigned party would first isolate the subject on-film by physically blackening its surroundings. And then, composite it onto multiple background films via an optical printer, and create a new copy with the added effects. This “overload” of effects and frequent reprinting was also one of the many reasons why older films used to be extremely grainy. The more the effects, the more the grittiness.

Additionally, movies also implemented the miniature effect, where hand-crafted scale models were used to recreate realistic locations, objects and even monstrous creatures. Movements were controlled either by concealed puppeteering or through stop-motion animation (example: King Kong (1933)). Lighting for model houses and streets were achieved through switch-activated LED bulbs, while vehicles were basically kids’

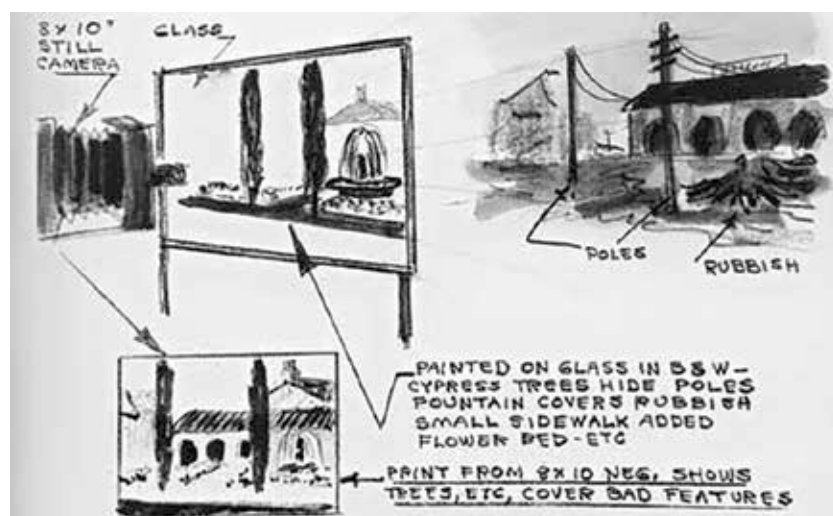


Diagram illustrating the workings of a glass matte painting